

Creative Industries Course — Curriculum Dissection & Interactive Learning Design

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Scope and how to read this

This is a first-pass dissection of the six-track curriculum: 6 tracks, roughly 40 modules and about 200 named units. It works top-down: a reusable framework for turning any unit into an interactive experience (sections 3–4), then a track-by-track map of what each module must research, ask, teach and build (sections 5–9), then what the curriculum itself needs fixed before production (section 10).

Depth is deliberately uneven. Track 01 and Track 02 are dissected to unit level, because they are the foundation and the operating system that everything else references, and because your own examples (interactive timeline, light-and-colour tool) sit there. Tracks 03–06 are dissected to module level with unit notes; each of those tracks deserves its own pass at the same depth as Track 01 once the framework is agreed.

Every module entry follows the same five fields so the whole document can later be exported as a production brief:

1. **Core questions** — what the module must be able to answer for the learner
2. **Research needs** — what has to be sourced, verified or built before content can be written
3. **Objectives and learning outcomes** — objectives are what the course intends; outcomes are what the learner can demonstrably do afterwards, written as verbs
4. **Methodology** — how the learning actually happens
5. **Interactive format** — the primary element type from the taxonomy in section 3, plus the specific tool concept

Nothing in this document relies on prior conversations or outside assumptions; where a fact was looked up it is linked, and where a judgement is mine it is phrased as one.

Benchmarks: what "interactive" has to mean here

The honest benchmark finding is that no existing platform does what this curriculum needs. The creative-course market (Domestika, Skillshare, MasterClass-style) is high-production video with a project at the end; the interactive-learning market (Brilliant, the explorable-explanations community) is strongest in maths, physics and systems, not in cultural theory, material practice or studio critique. This course sits in the gap between them, so the benchmark is a set of borrowed principles, not a product to copy.

Model	What it does well	What to borrow	What it cannot do for this course
Explorable explanations (Bret Victor's reactive documents, Nicky Case, the explorabl.es hub)	The reader changes the author's assumptions and sees consequences; abstract ideas become concrete through play	The "reactive document" as the default page format: prose with live parameters embedded, not video with a quiz after	Works for systems with a clear model (light, gravity, grids); weak for contested history and ethics, which have no single model
Brilliant-style guided problem sequences	Progressive difficulty, immediate feedback, learner acts before being told	The rhythm: predict, act, observe, then read the explanation	Assumes a right answer; most of Track 01 and all of Track 03 have defensible answers, not correct ones
Domestika / Skillshare	Production quality, taught by practitioners, one portfolio project per course	Practitioner voice and a tangible project per module	Passive; no critique, no research method, no systemic thinking
Studio education (architecture, fashion, art school)	Critique, iteration, the desk crit, the pin-up, learning from peers' failures	The critique loop and the pin-up as first-class interactive elements, not add-ons	Not online, not scalable, dependent on a live tutor

Model	What it does well	What to borrow	What it cannot do for this course
Museum and archive interfaces (network-style timelines, non-linear time scales, counterfactual branching)	Non-linear time, connections between events, the same period read from several centres	The history units should be networks and maps, never a single line of dates	Usually curated one way; this course needs the learner to re-centre the map themselves
Simulation and digital-twin tools (PBR renderers, CLO3D, CFD, Unreal)	Ground truth of physics; consequences are real, not illustrated	Simplified in-browser versions as the teaching tools for light, drape, thermal mass, structure	Professional tools are too heavy to teach with; the course needs pedagogical models that keep the truth and drop the complexity

Two current developments are worth designing around. Research on evaluating explorables now formalises interactivity as a set of learner-controllable states and transitions, a useful rubric for judging whether a tool is interactive or merely animated ([EE-Eval, AIED 2026](#)). And generative UI is being used to produce guided, scaffolded simulations per topic with progressive challenge levels ([Google Research, Sept 2026](#)), which suggests the tools in this course could be built from a small set of authored engines plus generated variations rather than 200 hand-built pieces.

The working definition for this course: a unit is interactive when the learner must make a decision that changes what they see, and the change teaches something the prose alone could not. Clicking "next", watching an animation, or answering a multiple-choice question does not qualify.

The Unit Anatomy: one framework, twelve element types

Every unit in the curriculum can be built from the same eight-beat sequence and a palette of twelve interactive element types. Standardising this is what makes 200 units buildable: the engines are authored once, the content varies. It also gives the course a recognisable rhythm, the way a studio day has a rhythm.

The eight beats of a unit

Beat	What happens	Purpose	Typical share of unit time
1. Provocation	One artefact, image, sound or question with a visible tension; the learner records a first reaction before any teaching	Create a stake; capture a prior to compare against later	5%
2. Situate	A reactive-document brief: prose with live parameters, definitions on hover, regional and global examples side by side	Give the minimum context needed to act	15%
3. Manipulate	The unit's primary tool (from the taxonomy below); the learner changes variables and sees consequences	Build intuition through decisions	30%
4. Evidence	A case dossier of 3–6 real projects, each with provenance, region, date, and the specific decision that illustrates the unit	Show the concept surviving contact with reality	15%
5. Counter-position	The strongest critique of the unit's own premise, or a case where the principle fails or was misused	Prevent doctrine; model critical practice	5%
6. Make	A bounded exercise (20–60 min) that produces something: a map, a rule, a drawing, a spec line, a paragraph	Externalise thinking into an artefact	20%

Beat	What happens	Purpose	Typical share of unit time
7. Critique	The artefact goes to a pin-up: rubric-based self-review, peer review, or AI critique, with the rubric visible from the start	Learn to defend and to judge	5%
8. Transfer	Re-apply the unit's idea to an unfamiliar scale, medium or region in one move	Prove the knowledge is systemic, not local	5%

Beats 1, 3, 6 and 7 are non-negotiable in every unit. Beats 4 and 5 can merge in short units. The Make artefact from every unit is written into a persistent learner object, the **Concept DNA Notebook**, which becomes the working document of Track 02 and the dossier of Track 06; without that thread the course is a collection of lessons rather than a practice.

The twelve element types

Type	What the learner does	Best for	Example units
Atlas	Navigates a network or map of events, people, movements and places; re-centres it on a region; toggles counter-narratives on and off	Histories, lineages, influence, trade routes	1.1 all units; 3.1.1; 3.2.1; 3.3.1; 3.4.1
Simulator	Changes physically true variables (light position, chroma, grain direction, thermal mass, load) and watches the model respond	Anything with a physical or perceptual model	1.3 colour and light; 3.1.4; 3.2.6; 3.4.3; 4.8; 4.9

Type	What the learner does	Best for	Example units
Lens	Views the same artefact through switchable interpretive frames (class, gender, climate, labour, semiotics) and sees what each frame makes visible	Critical and cultural readings where there is no single answer	1.1.4, 1.1.5, 1.4.3, 3.4.6
Dissector	Peels an artefact apart layer by layer with annotations: seams, joints, assemblies, MEP, BOM	Construction, tectonics, anatomy of a thing	1.4.1, 3.2.5, 3.3.6, 3.4.4, 6.2
Scenario engine	Sets variables about the future or the site and sees a branching set of consequences; can rewind and fork	Speculative futures, feasibility, risk, deal modelling	1.5.4, 2.1 speculative research, 3.2.9, 5.2
Rule engine	Writes constants, variables, thresholds and anti-rules, then tests candidate outputs as belongs / edge / failure / seductive-but-off	The Concept DNA itself and every grammar	2.2 entire module, 2.3, 3.4.8
Case dossier	Reads evidence cards with provenance and filters by region, decade, medium, decision type	Beat 4 in every unit	Universal
Pin-up	Posts an artefact; receives structured critique against the unit rubric; critiques others	Beat 7 in every unit	Universal
Calibrator	Perceptual drills: judge visual weight, kerning, chroma difference, grip size, acoustic decay, then see the measured answer	Training the eye, hand and ear	1.2.4, 1.3.2, 3.3.2, 4.1

Type	What the learner does	Best for	Example units
Ledger	Fills a live spreadsheet-like model (BOM, LCA, cash flow, FAR, tech-pack) and sees what a design change does to the numbers	The economic and ecological reality units	3.3.9, 3.3.10, 3.2.9, 5.2, 5.9
Notation studio	Draws, annotates or diagrams directly on the page: orthographic, storyboard, kinetic notation, technical flat	Track 04.1, tech-packs, movement	4.1, 3.2.10, 3.4.11
Field task	Leaves the screen: observe, measure, photograph, interview, then uploads evidence that the next beat uses	Ergonomics, site reading, vernacular, sound	1.3.1, 3.1.3, 3.2.3, 2.1 primary research

Three cross-cutting design rules follow from this:

- **The regional spine is structural, not a section.** The Atlas, the Case dossier and the Lens must all default to the Gulf, MENA and SWANA view with the global view one toggle away, rather than the reverse. That is the only way "regional centering" survives production.
- **Bilingual by architecture.** Arabic and English are both first languages of the interface, the glossary and the case dossiers. Unit 1.4.4 and 5.7 teach bilingual communication; the platform itself must demonstrate it.
- **Every tool exposes its model.** A simulator that hides its assumptions teaches deference. Each tool should have a "show the model" layer that reveals what is being computed and where the simplification lies, which is itself the content of 4.8.4 (ethics of visual representation) and 4.11.3 (algorithmic bias).

The research protocol: how the knowledge gets built

The curriculum needs four different kinds of knowledge, and each has a different research route, a different verification standard and a different failure mode. Treating them as one "research" job is the most likely way the content ends up generic.

Knowledge type	What it is	Research route	Verification standard	Failure mode if skipped
Canonical theory	The texts and frameworks a unit rests on (phenomenology, semiotics, relational aesthetics, cybernetics)	Literature review; primary texts read, not summaries of summaries; secondary literature for the critiques	Every claim traceable to a named text and page; the counter-position sourced too	The unit repeats the Wikipedia version and a practitioner reader spots it instantly
Physical and perceptual models	The truth behind every simulator: colour appearance, light transport, fabric mechanics, thermal behaviour, structural load, acoustics	Engineering and perception literature; existing open simulators as reference implementations; a domain reviewer per model	The tool's model checked by someone who works with the real thing; simplifications documented in the "show the model" layer	A beautiful tool that teaches something false
Case evidence	The 3–6 projects per unit with the specific decision that illustrates it	Project documentation, interviews with practitioners, archives, site visits; regional cases sourced first and directly	Provenance on every card: who, where, when, source; image rights cleared; the practitioner's own account preferred over press	Cases drift toward the same twenty famous Western projects

Knowledge type	What it is	Research route	Verification standard	Failure mode if skipped
Regional and vernacular knowledge	Gulf, MENA, SWANA histories, crafts, garments, buildings, oral histories, taxonomies	Archives and institutions in the region, oral history interviews, craft practitioners, regional academics; primary fieldwork	Cultural authority: a named consultant or community source per body of knowledge, credited in the unit; nothing presented as "traditional" without a source and a date	Extraction: exactly the practice unit 1.1.4 warns against, committed by the course itself

The research questions every unit must answer before writing

These are the same for all 200 units; the answers are what the curriculum author, a researcher or an agent team produces per unit.

1. What is the one idea this unit exists to make the learner understand, in one sentence?
2. What does the learner most likely believe before the unit, and what is the specific misconception the Provocation should expose?
3. What is the model — physical, historical, economic, semiotic — that the Manipulate tool will let them play with, and what are its three to six controllable variables?
4. Which three regional cases and which three global cases demonstrate the idea, and what exact decision in each is the evidence?
5. What is the strongest argument against the unit's premise, and who has made it?
6. What does the learner make, and what rubric will it be critiqued against?
7. What unfamiliar context proves transfer?
8. Which later units depend on this one, and which earlier units it depends on?
9. Who is the cultural authority or domain reviewer for this unit's content?
10. What terminology needs a bilingual glossary entry?

Sourcing tiers

- **Tier 1** (must be cited in the unit): primary texts, the practitioner's own documentation, archival records, peer-reviewed models, direct interviews.
- **Tier 2** (may inform, must be corroborated): monographs, institutional publications, established secondary literature.
- **Tier 3** (never load-bearing): press, blogs, marketing material, AI-generated summaries.

Production sequence per module

Research the whole module before writing any unit, because the case dossier and glossary are shared across its units. Then per unit: answer the ten questions, build or configure the tool, write the reactive brief around the tool, assemble the dossier, write the Make brief and rubric, and only then record any video. Video last, because it is the most expensive thing to change.

Track 01 — Context & Core Fundamentals

Track 01's job is to replace intuition with defensible judgement across five domains: history, form, the senses, material and language, and systems. It is also where the two shared spines of the course are born: the regional Atlas (from 1.1) and the Concept DNA Notebook (first entries from 1.2 onward). The track should be assessed not by recall but by whether the learner can situate, read and defend a creative decision using each domain's vocabulary.

Module 1.1 — Historical, Cultural & Critical Frameworks

Purpose. To make history usable: a web of connected, contested trajectories the learner can re-centre on their own region and interrogate for power, labour and ethics.

Core questions. Why does design history look the way it does, and who wrote it? What was happening in the Gulf, Egypt and the northern Arab world at the moment each canonical Western movement was declared? When is borrowing reference, quotation, translation or appropriation, and who decides? Whose labour is missing from the credit line of any famous object?

Research needs. A cross-disciplinary event and movement dataset (art, design, architecture, fashion, film, performance, digital) with dates, places, people and typed relations (influenced, reacted against, traded with, colonised, funded). A parallel regional dataset for the Gulf, Egypt, Levant, Maghreb and SWANA to the same resolution, built

with regional historians and archives, not inferred from Western timelines. A documented ethics framework (reference to appropriation spectrum) with a case bank of disputed borrowings. Oral histories and craft lineages recorded with named consultants. This module has the highest research cost in the course and should be started first.

Objectives. Establish critical historiography as a working tool; anchor the course regionally; install the ethics of practice before any making happens.

Learning outcomes. The learner can: place any artefact in at least two intersecting trajectories, one regional and one global; identify the dominant narrative and one counter-narrative for a given movement; classify a borrowing on the reference-appropriation spectrum and say what consultation it would require; produce a credit map for a canonical object naming obscured contributors.

Methodology. Atlas-first. The learner enters through the network, not through a lecture, and every unit is a different way of reading and editing the same Atlas. Cases come from the dossier; the Make artefacts are additions to the learner's own layer of the Atlas.

Interactive format. One Atlas engine with four modes, plus a Lens tool for the ethics units.

Unit	Core question	Tool concept	Make artefact
1.1.1 Interdisciplinary Global Discourse	What does a single decade look like across eight creative fields at once?	Atlas in "slice" mode: pick a year range and see all disciplines' events stacked, with cross-field links highlighted	A one-decade cross-disciplinary slice with three links the learner argues are causal
1.1.2 Critical Historiography & Counter-Narratives	Which lines on the map were drawn after the fact, and by whom?	Atlas in "authorship" mode: every canonical narrative shows its author, date and institution; a toggle overlays counter-histories and dims the canon	A re-drawn trajectory for one movement with the canonical and counter versions side by side

Unit	Core question	Tool concept	Make artefact
1.1.3 Vernacular Intelligence & Regional Taxonomies	What systemic knowledge is embedded in a vernacular object, story or term?	Atlas in "taxonomy" mode: regional objects, materials, terms and tales as nodes; a Field task to collect one item locally	A taxonomy card set for one local vernacular system, sourced and credited
1.1.4 Power Dynamics & Ethics of Practice	Where on the spectrum from reference to appropriation does this act sit?	Lens tool: a case of borrowing read through consultation, attribution, power asymmetry and economic flow; the learner places it on the spectrum and the tool shows how practitioners and affected communities placed it	A written ethics position for one disputed case, with the consultation it would have required
1.1.5 Authorship, Representation & Obscured Labour	Who actually made this, and why is only one name attached?	Dissector on a canonical object showing its credit network: designer, workshop, technicians, communities, suppliers, with named and unnamed nodes	A full credit map for one object, with sources for each contributor

Unit	Core question	Tool concept	Make artefact
1.1.6 Regional Centering & Geopolitical Specificity	What does the history of design look like when the Gulf is the centre of the map?	Atlas in "re-centre" mode: the projection recomputes around Dubai, Cairo or Beirut; distances and densities change	A regional narrative for one period written as the primary generator, not the periphery

Module 1.2 — Morphology, Typology & Formal Logic

Purpose. To give the learner control over form itself, independent of medium, so that later studios share one formal vocabulary.

Core questions. What survives when an object changes scale by a thousand? What is the structural DNA of an archetype, and how far can it be bent before it stops being that thing? What invisible geometry is holding a composition together? Where is the visual weight, and why does the eye go there?

Research needs. A cross-scale formal vocabulary reconciled across graphic design, product, interiors, architecture and interface. A typology catalogue (chair, garment, room, interface, vessel) with the invariant features of each documented. Reference literature on grids, proportion systems and gestalt perception; a perception model for visual weight good enough to power a Calibrator.

Objectives. Make form a system of decisions; establish scale-agnostic thinking; give the learner a repeatable method for subverting archetypes.

Learning outcomes. The learner can: describe a form's positive and negative structure at two scales; name an archetype's invariants and produce a variant that breaks three expected features while keeping identity; construct and justify a generative grid; predict and then correct the visual weight of a composition.

Methodology. Simulator and Calibrator heavy. Learn by manipulating form, testing perception, then writing the first formal rules into the Concept DNA Notebook.

Unit	Core question	Tool concept	Make artefact
1.2.1 Dimensionality & Scale Agnosticism	What changes and what holds when this form is a logo, a chair and a building?	Simulator: one form rendered at three scales with a positive/negative toggle; the learner edits the void and sees the object respond	A form studied at three scales with notes on what held
1.2.2 Typological Subversion	How much of a chair can I remove before it is no longer a chair?	Rule engine on an archetype: invariant features listed; the learner switches features off and the tool renders and asks "still a chair?" with practitioner cases at each threshold	One subverted archetype with its three broken expectations and preserved identity stated
1.2.3 Structural Skeletons & Generative Grids	What hidden geometry is governing this?	Simulator: overlay grids on real works; then set a modular ratio and watch a layout and a facade regenerate from it	A generative grid with three outputs at different scales
1.2.4 Visual Gravity, Asymmetry & Tension	Where does the eye go first, and where does it rest?	Calibrator: mark the perceived centre of weight; the tool reveals the measured centre and the fixation model; then the learner rebalances a composition to a target tension	A composition with intentional asymmetry and a written account of its tension

Module 1.3 — Somatic, Optical & Interaction Dynamics

Purpose. To move from static order into the lived dimension: the body, the eye, the hand and time.

Core questions. What does the body know that the drawing does not? How does one colour change another next to it, and how does a culture change both? How does a thing tell you how to use it without a label? What is the sequence of psychological states a person passes through from approach to departure?

Research needs. Anthropometric and cognitive-load data usable for a teaching model. A colour-appearance and light-transport model faithful enough for a browser Simulator (simultaneous contrast, chroma, illuminant, surface reflectance) with a domain reviewer. A cross-cultural colour-coding dataset with regional depth. Affordance and haptic literature; a catalogue of micro-interactions with measured feedback. Case studies of choreographed spatial sequences, including regional ones (courtyard entry, souq threshold, majlis approach).

Objectives. Install the body as the metric; make colour and light systemic rather than decorative; teach time as a design material.

Learning outcomes. The learner can: diagnose a design against three somatic constraints; build a palette and lighting logic that produces a stated hierarchy and depth; specify the feedback of a micro-interaction in physical terms; storyboard a threshold sequence with named psychological states.

Methodology. This is the module you described: build the environment, then let the learner learn by moving the lights. The Simulator is the lesson; the prose sits around it. Field tasks bring the body in.

Unit	Core question	Tool concept	Make artefact
1.3.1 Ergonomics & Somatic Truths	What does this design ask my body to do, and can it?	Simulator: a posable body model interacting with an object or interface; sliders for reach, grip, posture, load; red zones appear where strain or cognitive load exceeds thresholds; a Field task to measure one real object	A somatic audit of one object with three constraint violations found

Unit	Core question	Tool concept	Make artefact
1.3.2 Chromatic Systems & Optical Physics	Why does the same colour look different here, and what should I do about it?	Simulator: a room with movable lights of adjustable temperature and intensity, swatch surfaces with adjustable reflectance and chroma; a "perception" layer showing simultaneous contrast and depth cues; a cultural-coding overlay by region	A systemic palette plus a lighting logic that achieves a stated hierarchy, tested in the tool
1.3.3 Affordance & Micro-Interactions	How does it tell me what to do?	Calibrator plus Dissector: the learner predicts how an unlabelled object or UI is used, then sees the mechanism; a resistance/feedback curve editor for handles, switches and digital states	A micro-interaction spec (weight, resistance, sound, haptic) for one physical and one digital state change
1.3.4 Sequence, Duration & State Changes	What happens to a person between the street and the seat?	Notation studio with a time scrubber: storyboard a journey with compression, reveal and threshold beats; the tool plots the intended emotional curve	A choreographed sequence for one space with a named state at each beat

Module 1.4 — Materiality, Semiotics & Textual Systems

Purpose. To reposition material, surface, symbol and text from finishes to structure.

Core questions. What does this material say, where did it come from, and how does it behave under load, heat and time? How fast does the eye move across this surface, and what will it look like in ten years? What does this shape, colour or gesture mean here, and what does it mean elsewhere? Is the text part of the thing or stuck on afterwards?

Research needs. A materials library with behavioural data (tensile, thermal, drape, reflectivity), lifecycle and sourcing ethics, and regional materials documented with craft sources. A patina and ageing image dataset over time. A semiotics case bank with regional symbols and their contested readings. Bilingual typographic reference, including Arabic type systems and bidirectional layout conventions.

Objectives. Teach material as text and as physics; make ageing a design decision; make language a structural material in both scripts.

Learning outcomes. The learner can: justify a material choice on behavioural, ethical and cultural grounds; predict a surface's ageing and design for it; decode and deploy a symbol with awareness of its regional readings; integrate bilingual text into a form so it cannot be removed without breaking the design.

Methodology. Dissector and Simulator for the physical units, Lens for semiotics, Notation studio for typography.

Unit	Core question	Tool concept	Make artefact
1.4.1 Material Epistemology & Behaviour	What is this material for, and what is it doing to the concept?	Dissector plus Simulator: a material card opens into behaviour tests (bend, heat, wet, load) and a supply-chain map; swap materials in a reference object and watch behaviour and ethics scores change	A material argument for one object: three candidates, one chosen, on evidence
1.4.2 Surface Grammars & Patina	How will it look in ten years, and is that what I want?	Simulator: adjust finish, reflectivity and friction; scrub a time slider to see wear, oxidation and decay under stated use	A surface specification with a designed ageing trajectory

Unit	Core question	Tool concept	Make artefact
1.4.3 Semiotics & Signifiers	What does this mean, and to whom?	Lens: one artefact read through regional, generational and disciplinary frames; associations appear and conflict	A semiotic map of one design choice with its conflicting readings
1.4.4 Typographical & Textual Space	Is the text load-bearing?	Notation studio: place Arabic and English text into an object or space; a bidirectional layout engine and a "remove the text" test shows whether the design survives	One bilingual textual composition where removing the text breaks the design

Module 1.5 — Systems Thinking & Planetary Scale

Purpose. To make every later design decision answerable to its systemic and planetary consequences.

Core questions. What does this object disrupt when it enters the world? Where is the machine in this system, and what does it want? What happens to this thing after it is released, and who pays for that? How does this world behave when the future changes?

Research needs. System-mapping methodology and a case bank of interventions with documented second-order effects. Literature on cybernetics and on algorithmic agents as actors. Lifecycle data (embodied carbon, water, toxicity, repairability) for representative products, spaces and digital outputs, with regional supply-chain specifics. Futures methodology and a set of regional scenario drivers (climate, energy transition, demographics, technology).

Objectives. Move the learner from object to system; introduce non-human actors as design conditions; make lifecycle and futures first-class.

Learning outcomes. The learner can: map an intervention's ecosystem and identify three second-order effects; describe an algorithmic actor's role in a designed environment; estimate a design's ecological afterlife and redesign one decision to reduce it; run a scenario and state how the design would need to adapt.

Methodology. Scenario engine and Ledger; this module previews the tools Track 04.9 and Track 06.3 use at full depth.

Unit	Core question	Tool concept	Make artefact
1.5.1 Complexity & Ecosystem Mapping	What does this connect to that I have not drawn?	Scenario engine in mapping mode: drop an intervention into a system and watch links propagate; hidden actors reveal on inspection	An ecosystem map for one intervention with second-order effects named
1.5.2 Cybernetics & Non-Human Actors	Who or what is also designing this space?	Simulator: a space with an algorithmic agent (feed, sensor, pricing engine) as a node; change its objective and see the environment change	A brief describing one non-human actor as a design condition
1.5.3 Lifecycle, Attrition & Ecological Weight	What is the true weight of this thing?	Ledger: a lightweight LCA where material, sourcing and repairability choices move the totals in real time	An ecological ledger for one object with one decision changed and its effect shown
1.5.4 Speculative Futures & Scenario Generation	What breaks first when the world shifts?	Scenario engine: set geopolitical, technological and climatic variables; branch and rewind; the design's assumptions are flagged as they fail	A scenario report: three futures, the design's failure point in each, one adaptation

Track 02 — Generative World Systems

Track 02 is not a set of topics; it is a single project run through four phases, and it should be taught as one. The learner enters with the Concept DNA Notebook from Track 01 and leaves with a versioned Concept DNA, a collection architecture and an anchor expression. The interactive centre of the whole course lives here: the **Rule engine** where the DNA is written, versioned and tested. Every other tool in Track 02 either feeds it or reads from it.

The defining design decision for this track is whether learners build the DNA for a world of their own choosing from Phase I, or first practise on a supplied "training world" and then build their own. The second is safer for an online cohort with no tutor at the desk, and the training world doubles as the course's own worked example; the first is more motivating. A hybrid — a supplied world for Phase I exercises, the learner's own world from Phase II — is my recommendation and the assumption below.

Module 2.1 — Phase I: Research, Narrative and World Premise

Purpose. Turn context into a creative position without declaring rules too early.

Core questions. What do I actually know, how do I know it, and what are the limits of that evidence? Where does data stop and insight begin? What is my worldview, and what productive contradiction sits inside it? Who is this for, and what should they feel? What does this world enable, protect, resist and reject?

Research needs. A research-methods reference covering primary, secondary, archival and speculative routes with regional examples of each (a fieldwork case in a souq, an archival case from a Gulf institution). A source-credibility framework. Worked examples of the "insight threshold": research clusters and the leap that came from them, ideally from practitioners' own notebooks. Narrative-structure references from literature, film, sound and performance, with archetype and ritual taxonomies. A training world, fully documented, for the exercises.

Objectives. Install methodological rigour; teach synthesis as a skill, not a mystery; produce a defensible World Premise.

Learning outcomes. The learner can: plan and execute a mixed-method research sprint and grade its sources; cluster findings and articulate one insight that is not a restatement of the data; write a positioning statement with a named central contradiction; define the world's agency and perimeter as explicit enables/resists/rejects lists with non-goals.

Methodology. A guided research sprint on the training world, then the same sprint on the learner's own world. Beats 3 and 6 dominate; the Pin-up critiques premises, not aesthetics.

Unit	Core question	Tool concept	Make artefact
2.1.1 Comprehensive Research Methodologies	Which method answers which kind of question?	Scenario engine as a research planner: choose methods for a stated question; the tool shows what each will and will not reveal, with time and access costs; Field task to run one primary method	A research plan with methods matched to questions
2.1.2 Epistemological Rigour & Boundaries	How much can I trust this, and what am I not seeing?	Calibrator: grade sources blind, then see expert grades and the bias each source carries	A graded source register with stated limits of evidence
2.1.3 Synthesis & the Insight Threshold	What is the leap, and can I make it on purpose?	An interactive clustering board: drag findings into clusters, name patterns; the tool withholds "insight" status until the statement is not present in any single source	One insight statement with the cluster trail behind it
2.1.4 Philosophical Stance & Positioning	What do I believe, and what contradicts it inside my own position?	Lens: read the emerging world through five worldviews; a contradiction finder flags where the position is flat	A positioning statement with its central contradiction
2.1.5 Human & Experiential Intention	Who is inside this world, and what must they feel?	Scenario engine in persona mode: build scenarios, not personas; test each experiential outcome against them	Three human scenarios with specific experiential outcomes
2.1.6 Narrative & Semiotic Architecture	What are the stakes, the archetypes and the rituals?	Atlas of narrative structures from literature, film, sound and performance; drag structures onto the world and see which archetypes it demands	A narrative architecture: stakes, archetypes, defining rituals
2.1.7 Systemic Agency & Perimeter Setting	What does this world refuse to be?	Rule engine, first use: write enables/protects/resists/rejects/transforms and non-goals; the tool generates hypotheses to test in Phase II	The World Premise v0.1 with perimeter and hypotheses

Module 2.2 — Phase II: Concept DNA and Generative Systems

Purpose. Turn the premise into a scalable, versioned operating system that others can execute from.

Core questions. What is immutable, what is flexible, and where exactly does it break? Which grammars does this world need, and how specific must each be? How does this DNA relate to the practice, the collection and the masterbrand? How do I know a seductive output is off-DNA? Can a stranger, or a machine, build from this without me?

Research needs. Precedents of published design systems, brand codes and world bibles from studios, games and film, analysed for their constants/variables/thresholds structure. A grammar taxonomy (formal, material, construction, chromatic, spatial, atmospheric, sensory, behavioural, interaction, temporal, narrative, verbal) with exemplar rule statements at the right level of specificity. A brief-writing corpus: contributor briefs and AI prompt architectures that worked and failed. The training world's full DNA as the worked example.

Objectives. Make the DNA an executable document; teach thresholds as the real content of any system; make blind interpretation the test of quality.

Learning outcomes. The learner can: write a Concept DNA with immutable principles, anti-rules, constants, variables and numeric or descriptive thresholds; author all twelve grammars at a level a stranger can act on; map DNA to practice, collection and masterbrand; classify outputs as belongs / edge / failure / seductive-but-off and defend each; write a brief that a peer or an AI agent executes correctly on first pass.

Methodology. Rule-engine-centred. The learner writes the DNA, submits outputs against it, and — the key mechanic — hands the DNA to a blind executor (peer, or an AI agent inside the platform) and grades what comes back. The gap between intention and blind output is the lesson.

Unit	Core question	Tool concept	Make artefact
2.2.1 The Core Operating System	What must never change, and what must this never be?	Rule engine: principles, rules, anti-rules, anti-DNA; every rule tested against the training world's outputs	Concept DNA v1.0 core

Unit	Core question	Tool concept	Make artefact
2.2.2 Variables, Constants & Thresholds	Where is the edge?	Simulator: push a variable from constant toward threshold on a rendered output and mark the break point; compare with peers' marks	A threshold map for five variables
2.2.3 Comprehensive Systemic Grammars	What is the exact language of this world in each dimension?	A twelve-tab grammar editor; each grammar has a specificity meter and example outputs generated from it	Twelve grammars, each with one generated exemplar
2.2.4 Ecosystem Identity & Alignment	How does this DNA sit against practice, collection and masterbrand?	Atlas of DNAs: nested and overlapping rule sets; conflicts highlighted	An alignment map with resolved conflicts
2.2.5 Category Obligations & Systemic Flexibility	How different can two outputs be and still be one world?	Simulator: generate a family of outputs from the DNA; sliders for variation; a sameness/cloning detector	A variation study: five outputs, no clones
2.2.6 Boundary Testing Frameworks	Is this belongs, edge, failure or seductive-but-off?	Calibrator: sort a stream of outputs into the four bins; see practitioner sorting; disagreements become discussion	A tested set with reasoning per item
2.2.7 Execution & Blind Interpretation	Can someone else build it from this?	Pin-up in blind mode: the learner's brief goes to a peer or an AI agent; the returned output is scored against the DNA; the brief is revised	A revised brief and the before/after outputs

Module 2.3 — Phase III: World and Collection Architecture

Purpose. Design the ecosystem of outputs and their relationships before committing to a hero.

Core questions. What is this collection's thesis and where does it sit in the arc? What role does each output play? How do families, series, editions and drops relate? What holds together across scales? What can the audience absorb, and what can the studio afford to risk? What gets cut? Through which carriers does this world meet reality?

Research needs. Collection and range-planning models from fashion, furniture, product, publishing and games, with the role vocabulary (anchor, hero, core, entry, experimental, supporting) reconciled. Case studies of editorial rhythm and of cutting. Distribution and carrier taxonomies with regional retail and cultural-institution specifics. Cost and risk distribution examples from real portfolios (anonymised where needed).

Objectives. Teach architecture before expression; make editing a design act; connect the world to carriers.

Learning outcomes. The learner can: write a collection thesis positioned in a narrative arc; assign roles to every planned output; structure families, series and drops with rules; maintain cross-scalar continuity; balance complexity against accessibility, cost and risk; cut outputs and justify it; select carriers and interfaces for the world.

Methodology. The Rule engine gains a collection layer; the Ledger appears for cost and risk; the Pin-up critiques maps, not objects.

Unit	Core question	Tool concept	Make artefact
2.3.1 Architectural Thesis & Narrative Arc	What is this iteration for, in the story of the brand?	Atlas in arc mode: place the collection on the brand's timeline and see what it must advance	A collection thesis and arc position
2.3.2 Typologies & Hierarchical Roles	What is each thing's job?	Rule engine: a role assigner; the tool flags a collection with no anchor or too many heroes	A role map for every planned output

Unit	Core question	Tool concept	Make artefact
2.3.3 Structural Ecosystems	How do the parts relate and release?	A family tree editor with rules for modular, serial and limited configurations	A structural ecosystem diagram with configuration rules
2.3.4 Cross-Scalar Aesthetic Architecture	Does the detail belong to the environment?	Simulator: zoom from environment to artefact; continuity checks against the grammars	A cross-scalar continuity sheet
2.3.5 Strategic Distribution & Implementation Risk	What can we afford to get wrong?	Ledger: distribute cost and risk across the portfolio; see exposure change as outputs move roles	A risk and cost distribution map
2.3.6 Temporal Curation & Editorial Rhythm	What comes out when, and what never comes out?	A drop-sequencing timeline with an audience-attention model; a cutting-room mechanic where cuts must be justified	A sequenced release plan with a cut list
2.3.7 Public Carriers & Interfaces	How does this world touch reality?	Atlas of carriers (spatial, digital, print, event, retail) with regional options; select and justify	A carrier strategy

Module 2.4 — Phase IV: Applied World Construction and Development

Purpose. Build the anchor and a family through a large integrated brief, with the DNA and the outputs revising each other.

Core questions. Which practice is primary, and which adjacent translation stress-tests the system? How does an abstract rule become a joint, a seam or a screen? Who is using this, when, and how? What did prototyping reveal that the drawing hid? Which directions

were abandoned and why? Where does my competence end? What evidence proves this is resolved?

Research needs. Worked case studies of DNA-to-object translation, ideally with the studio's revision history visible. Prototyping and simulation method references appropriate to each practice (these are elaborated in Track 03 and Track 04, which this module should cross-link rather than duplicate). Specialist-collaboration case studies. An evidence-dossier template.

Objectives. Deliver a resolved anchor and family; make revision history visible and valued; teach the limits of solo practice.

Learning outcomes. The learner can: define a primary practice and execute one adjacent translation; translate DNA into technical decisions; ground outputs in scenarios and rituals; prototype, test and revise on evidence; document abandoned directions; identify specialist needs; compile a complete evidentiary output with an updated DNA.

Methodology. Studio mode. The platform becomes a project space: Pin-ups at milestones, a revision history that is graded, and the practice studios (Track 03) and tool studios (Track 04) available as reference. Critique is the main interactive element.

Unit	Core question	Tool concept	Make artefact
2.4.1 Practice Integration & Anchor Definition	What is the hero, and what is its translation?	Rule engine: declare practice, anchor and adjacent practice; the tool lists which grammars each will stress	An anchor and translation brief
2.4.2 Technical Translation & Construction Logic	How does the rule become a detail?	Dissector in reverse: from grammar statement to technical decision, with construction routes to choose	A technical route document
2.4.3 Contextualisation & Human Interaction	Who is this with, at 7am and at midnight?	Scenario engine: run the anchor through daily rituals and audience types	A scenario-grounded use document

Unit	Core question	Tool concept	Make artefact
2.4.4 Prototyping, Simulation & Constraint Negotiation	What did reality say?	Links to the practice's simulators (Track 04); a constraint log that records each negotiation	A prototype record with constraints negotiated
2.4.5 Iterative Evolution & Critique	What did I abandon and why?	Pin-up with a branching revision history; abandoned branches are kept and annotated	A revision history with abandoned directions
2.4.6 Specialist Collaboration	Where does my knowledge end?	Calibrator: rate your competence per required specialism; gaps become collaboration briefs	A specialist-needs map
2.4.7 Final Evidentiary Outputs	Is this ready for the world?	An evidence dossier assembler with a completeness check against the DNA	The Phase IV dossier and Concept DNA v2.0

Track 03 — The four practice studios

The four studios share one structure, and the curriculum already follows it: lineage, then conceptual method, then the medium's physics, then construction, then compliance and economics, then lifecycle. That regularity is an asset: one studio template with practice-specific tools. Each studio should also be entered through the same door — the learner's own Concept DNA from Track 02 — so the studio is a translation exercise rather than a fresh course. Below, each studio has its purpose, its shared research needs, and a module table; the studios are shown at module depth, and each deserves the Track 01 treatment in a second pass.

Across all four, the two heaviest research investments are a **materials-and-mechanics model library** (drape, load, thermal mass, light, sound) shared by every simulator, and a **regional practitioner interview programme**, because the lineage modules (3.1.1, 3.2.1, 3.3.1,

3.4.1) each promise regional depth that does not exist in the standard literature and cannot be assembled from it.

Studio 3.1 — Art, Installation & Experimental Practice

Purpose. Suspend utility to learn what space, time, light, sound and audience can do when the brief is a provocation. **Studio-specific research:** documentation of ephemeral works (the archive is often all that survives), acoustic and light-physics models for the simulators, regional mythologies and the desert sublime sourced from regional writers and oral tradition, and the economics of patronage in the region's institutions.

Module	Core question	Research needs	Primary format	Make artefact
3.1.1 Lineage, Theory & Mythos	What does the container do to the work, and what lineages let me subvert it?	Land Art, Light and Space, Situationism, institutional critique, relational theory; regional mythologies sourced directly	Atlas plus Lens (white cube / black box / site)	A lineage position for the learner's DNA
3.1.2 Conceptualisation & Generative Logic	Deductive or inductive, and what is the what-if?	Translational method precedents (data to form); constraint-driven ideation cases	Scenario engine as a what-if generator; a translation tool mapping non-spatial data to volume	Three what-if provocations, one developed
3.1.3 Spatial Disruption, Scale & Intervention	What does scale do to power?	Site-specific versus site-responsive cases; engineering of monumental and micro works	Simulator: a human figure in a scalable intervention; Field task on a real site	A site reading and one intervention at two scales

Module	Core question	Research needs	Primary format	Make artefact
3.1.4 Volumetric Chroma, Light & Matter	Can light be a wall?	Light-transport and haze models; Ganzfeld perception; altered-artefact and volatile-matter cases	Simulator: haze, projection, refraction, saturation; time-scrub for decay	A light-and-matter study with a stated perceptual effect
3.1.5 Sonic Topography & Acoustic Architecture	Can sound be a boundary?	Spatial audio, infrasound physiology, anechoic perception; site soundscape method	Simulator with binaural playback: place sources, set frequencies, walk the space	A sonic plan with a bodily effect specified
3.1.6 Synthetic Realities & Algorithmic Complicity	What happens when I give up control?	XR practice cases; live-data-driven works; non-human-audience precedents	Simulator: a live data feed drives an installation parameter; a post-human spectator mode	A generative rule set handing one variable to an external system
3.1.7 Relational Aesthetics & Durational Practice	What are the rules of engagement, and what remains?	Participatory-work rulebooks; documentation-as-artwork cases	Notation studio for audience journey; a decay timeline	Rules of engagement and an afterlife plan
3.1.8 Socio-Ecological Impact & Economics of Provocation	Who pays, what does it cost the planet, and whose city is it?	Right-to-the-city literature; ephemeral LCA; regional grant and patronage landscape	Ledger for ecological debt and funding; Lens on public/private space	A funding and circularity plan defending the DNA

Studio 3.2 — Architecture, Space & Experience

Purpose. Translate DNA and behavioural hypotheses into habitable, code-compliant, economically viable space. **Studio-specific research:** Gulf urbanism from courtyard to post-oil city with regional academics; climatic vernacular and passive-cooling physics for the simulators; regional building codes, life safety and zoning; commercial fit-out and lease realities from local developers.

Module	Core question	Research needs	Primary format	Make artefact
3.2.1 Lineage, Spatial Theory & Regional Modernisms	How is space felt, and what did the Gulf build before and after oil?	Phenomenology texts; typological lineages; Gulf urbanism sourced regionally	Atlas plus a Field task of a felt-space diary	A spatial-theory position
3.2.2 Programmatic Architecture & Brand Scenography	What is the internal logic before the form?	Hybrid typology cases; adjacency and zoning method; flagship scenography	Rule engine for programme; an adjacency-matrix tool	A programme and adjacency map from the DNA
3.2.3 Spatial Choreography & Narrative Journey	What does the body feel from approach to reveal?	Threshold and circulation cases, regional entry sequences	Simulator: a walkthrough with compression, expansion and reveal sliders; emotional curve plotted	A choreographed sequence
3.2.4 Adaptive Reuse & Spatial Palimpsest	What do I keep, strip and overwrite?	Heritage intervention cases; retrofit logistics; embodied-carbon argument	Dissector: a building as layers with a keep/strip/overwrite mode	A palimpsest strategy

Module	Core question	Research needs	Primary format	Make artefact
3.2.5 Tectonics, Fabrication & Assemblies	How do materials meet?	Joint and tolerance detailing; computational fabrication workflows; ageing	Dissector of details; Simulator for tolerance and weathering	A detail set with joints as expression
3.2.6 Atmospheric Engineering & Microclimate	Can I design what cannot be seen?	Solar geometry, thermal mass, evaporative cooling; daylight models; olfactory and acoustic design	Simulator: sun path, thermal mass, ventilation, daylight and reverberation	An atmospheric plan across a day and a year
3.2.7 Urban Interface & Civic Territory	Where does the building end?	Edge-condition cases; master-planning; regional zoning	Simulator of the street edge; a zoning-rule checker	An edge condition and civic position
3.2.8 Invisible Architecture	Can the back of house serve the concept?	Life safety and accessibility codes; MEP coordination	Dissector with services layers; an egress and access checker	A services and egress plan that survives the concept
3.2.9 Spatial Economics & Feasibility	Does it pay over ten years?	CapEx/OpEx models; FAR and yield; lease-line constraints	Ledger: fit-out economics with design changes moving totals	A feasibility ledger
3.2.10 Representation & Digital Twins	Can a contractor build this without asking?	Orthographic syntax; BIM and clash detection; specification schedules	Notation studio for plan, section, elevation; a digital-twin viewer	A drawing set and schedule extract

Studio 3.3 — Object, Product & Furniture

Purpose. Resolve discrete artefacts as tools, companions or provocations, through to manufacture and lifecycle. **Studio-specific research:** regional craft lineages recorded with artisans; a manufacturing-route library with cost and tolerance data; certification frameworks (BIFMA, CE, UL and regional equivalents); flat-pack and logistics models; repairability and end-of-life cases.

Module	Core question	Research needs	Primary format	Make artefact
3.3.1 Ontology, Critical Design & Craft Lineages	What does this object make people do?	Agency-of-things theory; critical design; regional craft interviews; emotional durability	Atlas plus Lens	An object position
3.3.2 Affordance, Ergonomics & Tactile Intimacy	Does the hand agree?	Anthropometric data; haptic and thermal-contact literature	Simulator: a hand model gripping adjustable forms; a resistance curve editor	An ergonomic and haptic spec
3.3.3 Typological Subversion & Proxemics	What does this object do to the room?	Archetype invariants; choreography of use; state-change mechanisms	Rule engine on archetypes; a room simulator showing circulation around the object	A subverted typology with its use ritual
3.3.4 Iterative Prototyping & the Physical Buck	What did the cardboard reveal?	Prototyping method; digital-to-analog loops	Field task (make a buck); Pin-up of prototypes; a tolerance test tool	A prototype log with failures

Module	Core question	Research needs	Primary format	Make artefact
3.3.5 Soft Architecture & Tensile Tectonics	How is comfort engineered?	Foam, spring and webbing data; pattern drafting for volume	Simulator: layered soft section with firmness, sink, recovery	A soft-goods section
3.3.6 Industrial Methodologies & Structural Honesty	Which route, and does the material tell the truth?	Manufacturing process library; joint and tolerance engineering	Dissector of joints; a route selector with cost and volume	A manufacturing route and joint spec
3.3.7 Compliance, Safety & Standardisation	Will it survive misuse and the law?	Load and fatigue standards; certification frameworks	Simulator for load testing; a compliance checklist	A test plan and compliance map
3.3.8 Cyber-Physical Hybridity	Where does the electronics live?	IoT integration; physical UI; parametric optimisation	Dissector of an intelligent object; a state-indicator designer	A hidden-architecture and state spec
3.3.9 Supply Chain Logic & Economics	Can it ship, and can it be sold?	Flat-pack cases; BOM and value engineering; packaging as threshold	Ledger: BOM with value-engineering moves; a nesting simulator	A BOM and packaging sequence
3.3.10 Circularity, Attrition & Lifecycle	How does it die well?	Right-to-repair; patina materials; end-of-life separation	Simulator for ageing; a disassembly sequence tool	A repair and end-of-life architecture

Studio 3.4 — Body, Fashion & Wearables

Purpose. Translate world-building to the moving body: soft architecture, textile physics, semiotics of dress, and the industrial and ethical pipeline. **Studio-specific research:** regional sartorial history (abaya, kandura, regional textiles) as climatic and cultural technologies, sourced from regional historians and makers; modesty as somatic architecture written with cultural authority; a textile-physics model for the drape simulator; grading mathematics; labour supply-chain documentation.

Module	Core question	Research needs	Primary format	Make artefact
3.4.1 Sartorial Lineage & Cultural Theory	What did silhouette do to power, and what did the region wear and why?	Western tailoring history; avant-garde deconstruction; subcultural trickle-up; regional dress and trade routes	Atlas with a silhouette-over-time layer; Lens on modesty	A sartorial position
3.4.2 Somatic Architecture & the Kinetic Envelope	Does it move with the body?	Ease calculations; kinetic garment cases	Simulator: an articulating avatar with adjustable ease; failure points shown in motion	A kinetic envelope study
3.4.3 Textile Epistemology & Material Performativity	What does the cloth want to do?	Woven versus knit mechanics; bias; bio-fabrication; embellishment as structure	Simulator: drape by grain, bias and structure; a bio-material library	A textile behaviour study
3.4.4 Pattern Tectonics & Construction	How does 3D become 2D and back?	Drafting and draping methods; seam architecture; closures	Notation studio: draft a pattern and see it fold onto the avatar; Dissector of seams	A pattern with a construction map

Module	Core question	Research needs	Primary format	Make artefact
3.4.5 Wearable Technology & Augmentation	Can the garment sense and act?	E-textiles; prosthetic augmentation; thermal engineering	Simulator: sensor placement and response; a microclimate model	A wearable-system spec
3.4.6 Semiotics of Dress & Subculture	What does this say, and to whom?	Dress codes, gender, class; regional vernacular readings	Lens on one garment through several frames	A semiotic reading and a subversion
3.4.7 Inclusive Matrix & Adaptive Design	Who is left out of the size chart?	Adaptive closures; seated drafting; non-linear grading	Simulator: grade a silhouette across a diverse body set and see distortion	A graded, adaptive silhouette
3.4.8 Collection Architecture & Merchandising	What sells the hero?	Hero/core/carry-over models; range planning	Rule engine for the matrix; a drop planner	A collection matrix
3.4.9 Digital Twins & XR Fashion	Can I design without cutting cloth?	3D simulation tools; digital-only garment economics	Simulator (a simplified drape engine); a digital-garment brief	A digital twin of one piece
3.4.10 Editorial, Styling & Performative Release	How does context change meaning?	Styling as design; art direction; runway as installation	Lens for styling variants; Notation studio for show choreography	A campaign and show plan
3.4.11 Industrial Translation & Tech-Pack	Can a factory build it without asking?	Technical flats; BOM and measurement charts; tolerance and shrinkage	Notation studio for flats; a tech-pack assembler with a QC tolerance calculator	A tech-pack

Module	Core question	Research needs	Primary format	Make artefact
3.4.12 Supply Chain Ethics & Circularity	Who made it, and what happens to it?	Labour auditing; regenerative sourcing; mono-material design	Atlas of a garment's supply chain; Ledger for material and labour impact	An ethical sourcing and end-of-use plan

Track 04 — Technical & Computational Studios

Track 04 has a structural problem worth naming before it is designed: it teaches tools the learner cannot run inside the course (Rhino, Unreal, DAWs, CNC, LIDAR), and tool-specific instruction goes stale within a year. The resolution is to teach the **thinking** of each tool with in-browser pedagogical versions, and to treat the professional software as a Field task with recommended paths rather than course content. Every Track 04 module then has two layers: the concept and its browser tool (built and owned by the course), and a "go and do it" layer that names current tools and can be updated without touching the unit. The strongest link to the rest of the course is that Track 04's browser tools are the same engines that Track 01 and Track 03 simulators run on, so the learner is finally shown the model they have been playing with.

Track-wide research needs. A survey of current tools per module, dated and scheduled for annual refresh; open-source reference implementations for each pedagogical engine (light transport, form-finding, drape, CFD-lite, LCA); a rendering-ethics case bank of misleading visualisations; an AI-workflow corpus of prompt architectures that held a DNA and ones that drifted.

Module	Core question	Research needs	Primary format	Make artefact
4.1 Analog Intelligence & Notational Logic	Can I see it, and can I draw what I see so someone else can build it?	Observational drawing pedagogy; orthographic conventions; kinetic notation systems	Notation studio with orthographic construction aids and a movement scrubber; Field task sketchbook	An orthographic set and a movement notation
4.2 Reality Capture & Reverse Engineering	How does the world become data?	LIDAR, photogrammetry, radiance-field and metrology methods with sample datasets	A point-cloud and mesh viewer with capture-quality diagnostics; Field task with a phone capture	A captured object or space, cleaned and audited
4.3 Algorithmic Geometry & Parametric Environments	What are the rules that generate the form?	Topology fundamentals; node-based logic; creative coding; form-finding physics	A browser node editor generating geometry; a form-finding simulator with gravity and tension	A parametric definition with three outputs
4.4 Synthetic Workflows & Machine Collaboration	Can an agent hold my DNA?	Agent-orchestration patterns; multimodal generation with constraints; failure cases	An agent-team designer; a constrained generation sandbox scored against the learner's DNA	A prompt architecture and a drift audit

Module	Core question	Research needs	Primary format	Make artefact
4.5 Spatial Computing, Simulation & Interface	How does a person move through the digital world?	Real-time engines; collision and physics; full-stack prototyping	A browser real-time scene with editable triggers and physics; an interface prototyper	A navigable prototype with behaviours
4.6 Information Architecture & Data Visualisation	Can this research be walked through?	Data parsing; visualisation grammar; spatialised data cases	A live visualisation builder on the learner's own research data; a spatialise mode	An interactive map of the learner's research
4.7 Physical Computing & Mechatronics	Can the object feel and move?	Microcontroller basics; sensor and actuator catalogues	A circuit and logic simulator with sensor inputs and actuator outputs; Field task with a starter kit	A sensing-and-acting behaviour spec
4.8 Optical Physics, Sonic Engineering & Visual Evidence	Is the render true?	PBR physics; cinematographic grammar; spatial audio; rendering-ethics cases	Simulator: PBR material and light lab with a "deception detector"; a spatial-audio composer	An honest render with a disclosure of its simplifications
4.9 Environmental Analytics & Lifecycle Computation	What does the maths say about the climate?	CFD and solar simulation principles; LCA databases and methods	Simulator: solar, wind and thermal on a form; Ledger: full LCA	A verified environmental performance report

Module	Core question	Research needs	Primary format	Make artefact
4.10 Digital Fabrication & Material Mechanics	Will the machine make what I drew?	CAM and toolpath logic; additive constraints; failure analysis	A toolpath previewer; a print-constraint checker; a failure simulator for thickness and shear	A fabrication-ready file with a failure analysis
4.11 Data Architecture, Cyber-Ethics & Version Control	Who owns the data, and which version is true?	BIM and PLM; version control for design; algorithmic bias and data sovereignty	A branching version history on the learner's DNA; a bias and sovereignty audit tool	A versioned DNA with a data-ethics statement

Tracks 05 and 06 — Implementation, Economics and the Capstone

Tracks 05 and 06 overlap substantially: the collaboration matrix, minimum viable implementation, release sequencing, IP and cultural authority, lifecycle operations, critical defence, systemic transfer and the professional dossier all appear in both. Read as written, a learner would meet the same content twice. The cleanest resolution is to make **Track 05 the strategic and economic layer** (brand, capital, law, studio OS, commerce, community, campaigns, expansion) and **Track 06 the capstone execution** of the learner's own project through handover, release, defence and dossier, with the duplicated units living only in 06. Section 10 lists the specific merges. The tables below assume that split.

The interactive character changes here. Fewer simulators; more Ledgers, Scenario engines and Pin-ups, because the content is decisions under financial, legal and human constraint. The most valuable single tool in these tracks is a **studio simulation**: the learner's project runs through a modelled year of cash flow, supplier delays, a funding negotiation and a plagiarism incident, and they must protect the DNA through each event.

Track-wide research needs. Regional funding landscape (state grants, cultural institutions, PPP structures) documented with named programmes and dated; regional IP and contract law with a lawyer's review; luxury and high-conversion commerce interface studies including Gulf platforms; community-brand case studies; Arabic-English campaign copy cases; import regulation and consumer behaviour for Gulf and Saudi markets; anonymised studio financials for the simulation.

Track 05 — Strategic Architecture and the Economics of Culture

Module	Core question	Research needs	Primary format	Make artefact
5.1 Masterbrand Strategy & Orbital Architecture	What houses the world, and how far can a sub-brand drift?	Brand-architecture models; endorsed and orbital cases	Rule engine extended to brand codes; an orbit-distance visualiser	A masterbrand and orbital framework
5.2 Capital Structuring & Creative Finance	Whose money, on what terms, and how long can we breathe?	Term-sheet mechanics; regional patronage and PPP; cash-flow modelling	Ledger: runway and burn with funding scenarios; a term-sheet dilution simulator	A funding strategy with a modelled runway
5.3 Legal Architecture, IP & Cultural Sovereignty	What can be taken from me, and what did I take?	IP filing routes by jurisdiction; contract templates; cultural-consultation frameworks	A protection-map tool by jurisdiction; Lens on a plagiarism and an appropriation case	An IP and consultation protocol

Module	Core question	Research needs	Primary format	Make artefact
5.4 Agency Operating System & Studio Governance	How does the studio run without me in every room?	Studio workflow and folder architectures; pitch structures; specialist vetting	A studio-OS builder; a pitch rehearsal Pin-up with structured challenge	A studio operating manual and a defended pitch
5.5 Omnichannel Commerce & Spatial Retail	Where does the transaction disappear?	Checkout and navigation studies of luxury and high-conversion platforms, regional included; O2O cases; flagship-as-media	Dissector of checkout flows; a Scenario engine linking online behaviour to store curation	An omnichannel journey map
5.6 Community Governance & Cult Brand Dynamics	How does an audience become a tribe without owning the brand?	Access-tiering cases; ritual and signifier design; co-creation governance	A tiering and access designer; a feedback-loop simulator	A community architecture
5.7 Narrative-Led Campaigns & Activation	Does the campaign belong to the world?	Campaign structures; bilingual copy cases; activation and registration workflows	Lens comparing Arabic and English copy for resonance; a journey mapper from awareness to conversion	A bilingual campaign and activation plan

Module	Core question	Research needs	Primary format	Make artefact
5.8 Implementation Logistics & Collaboration	What must give, and what must not?	Phased-drop planning; supplier dependency management; MVI cases	Scenario engine: a dependency timeline with slips; an MVI negotiator that flags non-negotiables from the DNA	An implementation plan (moves to 06.1 under the proposed split)
5.9 Venture Economics, Retail Curation & Expansion	Where next, at what margin, and at what total cost?	Regional import rules; consumer behaviour by market; COGS and margin structures; lifecycle costs	Ledger: margin architecture with sourcing substitutions; an expansion scenario by market	A pricing and expansion strategy
5.10 Transfer, Defence & Future Practice	Can the system survive a brief it has never seen?	Duplicate of 6.4–6.5	Merge into Track 06	—

Track 06 — Execution, Responsible Release and the Capstone

Module	Core question	Research needs	Primary format	Make artefact
6.1 Collaboration Ecosystem & Implementation Logistics	Who does what, by when, and what breaks first?	Briefing protocols; risk and conflict structures; drop sequencing	The studio simulation, year one: suppliers, slips, conflicts; MVI decisions logged against the DNA	A collaboration matrix and release sequence for the learner's project

Module	Core question	Research needs	Primary format	Make artefact
6.2 Technical Handovers & Regulatory Compliance	Can the world be built and released legally without me?	Handover package anatomy per practice; regulatory checklists; IP and credit frameworks	A handover assembler with a completeness audit; a compliance map; a credit and authority ledger	A full handover package
6.3 Responsible Release & Lifecycle Operations	What does it cost after launch, and how does it end?	Sourcing substitution cases; packaging and distribution; maintenance budgets; end-of-life routes	Ledger across the ten-year life; Scenario engine for supply substitution; a disassembly plan	A lifecycle operations plan
6.4 Systemic Transfer & Critical Defence	Does the knowledge belong to me, or to the projects I did?	Critique and defence methodology; transfer-task design	A blind transfer brief (unfamiliar medium, region and constraint) with a timed response; a recorded defence critiqued by peers and reviewers	A transfer response and a defended critique
6.5 Professional Dossier & Future Practice	What is the case, what is missing, and what comes next?	Casebook structures; competence frameworks; practice-planning models	A dossier builder drawing on every Make artefact since 1.1; a gap-analysis Calibrator; a practice-plan timeline	The studio casebook and a future-practice plan

Structural observations on the curriculum

The curriculum is unusually complete and unusually consistent in voice. The risks are the ones that come with that: duplication, scale, and a text register that will not survive contact with a learner unless the interactive layer carries it.

Duplications to resolve

Content	Appears in	Recommendation
Collaboration matrix, MVI, release sequencing	5.8 and 6.1	Keep in 6.1 as capstone execution; 5.8 becomes a short bridge or is removed
IP, patents, contracts, cultural sovereignty	5.3 and 6.2	5.3 teaches the law and strategy; 6.2 applies it to the learner's handover
Systemic transfer, critical defence, dossier, gap identification, future practice	5.10 and 6.4–6.5	Remove 5.10 entirely
Lifecycle, repair, end-of-life, packaging as threshold	1.5.3, 3.3.9–10, 3.4.12, 5.9, 6.3, 4.9	Keep 1.5.3 as the introduction, the studio versions as practice-specific, 4.9 as the computation, 6.3 as the operations; cross-link rather than re-teach
Typological subversion	1.2.2, 3.2.2, 3.3.3	1.2.2 teaches the method once; studios apply it to their archetype
Ergonomics and affordance	1.3.1, 1.3.3, 3.3.2, 3.4.2	Same pattern: method once, applied per studio
Regional lineage	1.1.6, 3.1.1, 3.2.1, 3.3.1, 3.4.1	One regional Atlas with practice-specific layers, rather than five separate histories
Tech-pack and handover documentation	3.2.10, 3.4.11, 6.2	One handover engine with practice templates
Digital twins	3.2.10, 3.4.9, 4.2, 4.5	Teach in 4.5, reference elsewhere

Scale and sequencing

- At roughly 200 units, even at a modest 90 minutes each, the course is around 300 hours of learner time. That is a degree, not a course. Either it is positioned as a multi-year programme with tracks sold and completed separately, or the studios (Track 03) are electives and a learner completes one or two, not four.
- Track 04 sits after Track 03 but Track 02 Phase IV and every Track 03 module need its tools. Either Track 04 runs in parallel as a toolbox opened on demand, or a compact "Track 04 essentials" is inserted before Track 02 Phase IV.
- The regional depth is promised in at least nine places and delivered by none of the standard literature. It has to be researched as a programme in its own right, with a budget, consultants and rights, before content writing begins, or every regional unit becomes a paragraph of intention.

Register

The curriculum text is written in the voice of a manifesto — "rigorous", "uncompromising", "brutal", "lethal misinterpretation" — which works as a positioning document and will read as intimidation inside a unit. The learner-facing register should be direct and warm, with the intensity carried by the tools and the critique, not the adjectives. The **Provocation** beat is where the manifesto voice belongs.

What is missing

- **Assessment and progression.** Nothing yet says how a learner passes, what a cohort is, or whether there is a certificate. The Concept DNA Notebook and the dossier can be the assessment spine, but the rubrics, the reviewer model (peer, AI, tutor) and the pass criteria need designing.
- **The training world.** Track 02 needs a fully worked example world, and it is the single most reusable asset in the course; it should be built early and used as the demonstration in every tool.
- **Sound, smell and touch in an online course.** Several units (3.1.5, 3.2.6, 3.3.2) teach senses a screen cannot deliver; Field tasks and shipped physical kits (material swatches, a starter electronics kit) should be a planned line item.
- **Onboarding and diagnostics.** A calibrating entry experience that places a learner and shows them the Atlas, the Notebook and the rhythm of a unit before Track 01 begins.

Open decisions and proposed next steps

Six decisions shape everything downstream; they are yours, and each has a default I would take if forced.

- ☐ **Programme shape.** One continuous course, or tracks as separately completable programmes with Track 03 studios as electives? Default: separately completable, studios elective.
- ☐ **Learner project.** Own world from Phase I, or training world first then own world? Default: hybrid, own world from Phase II.
- ☐ **Assessment and reviewer model.** Peer critique, AI critique, tutor critique, or a mix, and is there a credential? Default: rubric-based peer plus AI critique on every Make, tutor at four milestones, credential on the dossier.
- ☐ **Regional research programme.** Who leads it, which institutions and consultants, what budget and rights? No default; this is a hire and a partnership decision.
- ☐ **Engine count.** How many authored tools can be built and maintained? Default: twelve element types as engines, configured per unit, with generative variation only for exercises, never for the core models.
- ☐ **Physical kits.** Are material swatches and a starter electronics kit shipped to learners? Default: yes for the studios, no for Track 01.

Proposed sequence of work

1. Agree the framework in sections 3 and 4 and the merges in section 10.
2. Run the Track 01 dissection to production-brief level: the ten research questions answered per unit, the tool specs written, the case dossiers listed.
3. Build the Atlas engine and the light-and-colour Simulator first, as the two proofs of concept; they are the course's own examples and the two hardest engines.
4. Build the training world and the Rule engine together, since each tests the other.
5. Repeat the Track 01 dissection for Track 02, then one studio at a time.
6. Start the regional research programme in parallel with step 2; nothing regional can be written without it.

Sources

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